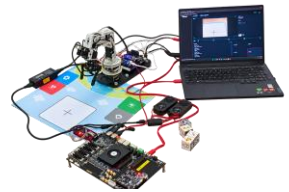
Xinyu SmartSort 芯语智分

A Voice-Visual Waste Sorting System Based on Zynq Hardware-Software Co-Design

Chongqing University Team
Chongqing University
重庆大学

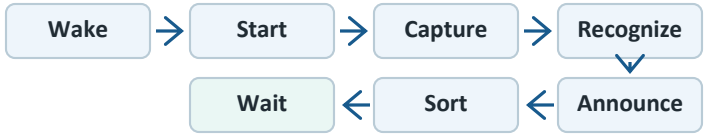


ALINX Z19 / Zynq UltraScale+ MPSoC

INTRODUCTION

Hear the command. Put sorting into action.

Xinyu SmartSort connects voice interaction, visual recognition and robotic action in a tabletop experience. One task turns a spoken command into a visible sorting sequence.



Typical application case

Campus sustainability activities.

Community outreach and interactive exhibits.

Embedded systems and FPGA teaching.

A visible lesson in greener living

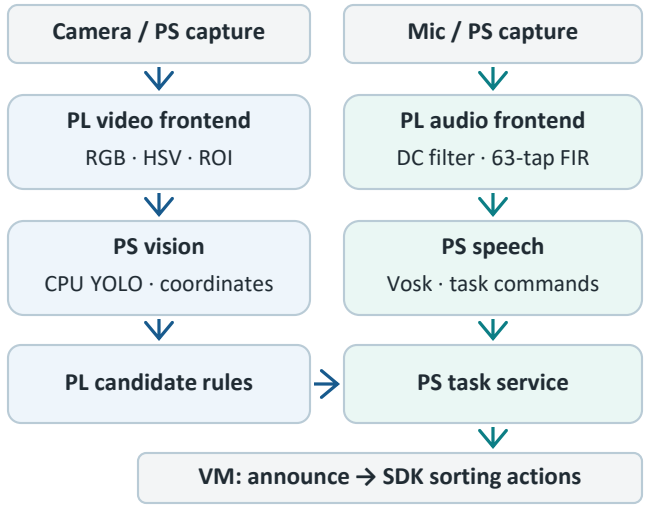
From “What is it?” to “Where should it go?”
Participants see recognition, feedback and action.



Object recognition example

One wake, one task

Wake “Xiaoya”, then start.
Announce the category before fixed-station sorting.
Return to waiting when done.



System Architecture (source profile)

CREATIVE DESIGN

PS/PL hardware-software co-design

1. PS runs YOLO, Vosk and task services.
2. PL processes image and audio streams:
   RGB/HSV, ROI and first moments;
   DC suppression and a 63-tap FIR.
3. Candidate rules link perception to one-task execution through the PS.

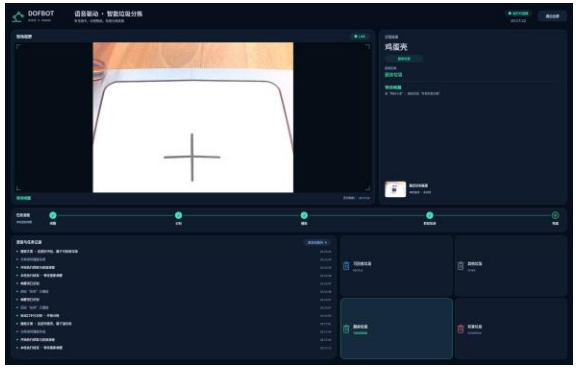
Interaction that people can follow

One wake-and-start cycle, one task.
Announce the category before grasping.
Modular design for teaching and reuse.

RESULT

Technical Specifications

Core: ALINX Z19 / Zynq UltraScale+ MPSoC
Structure: PS + PL + Ubuntu VM
Devices: camera, voice input, DOFBOT, host
Stack: Linux, Python, YOLO, Vosk
Modules: video, recognition, location, audio



VM / DOFBOT: visible task progress

Demonstrations & Deliverables

Voice-driven, fixed-station sorting demonstration.
Z19 PS/PL source modules and defined interfaces.
Next: joint integration and wider object coverage.